Feature-Driven Analysis of Online Purchase Intent Using SHAP and Predictive Models

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***Abstract*—In the current world, e-commerce is a very important business sector. This study integrates predictive modeling and SHAP (SHapley Additive exPlanations) to analyze factors influencing purchase intent in online environments. Using eight machine learning models, including XGBoost, SVM, and CatBoost, we identify CatBoost as the best-performing model with an accuracy of 89% and an F1-score of 67%. SHAP is then employed to interpret model outputs, revealing the importance and impact of key features on purchase intent. This approach bridges the gap between predictive performance and interpretability, offering actionable insights for personalized marketing, recommendation systems, and website optimization. The findings contribute to the growing field of explainable AI in e-commerce and provide practical strategies for enhancing user experience and driving business success.**

***Index Terms*—Online Purchase Intent, Machine Learning, SHAP, Explainable AI (XAI), Consumer Behavior.**

I. Introduction

The rapid proliferation of e-commerce platforms has transformed the way consumers shop - enabling a seamless, convenient and highly personalized experience. Understanding online purchase intent has become a critical area of research for businesses aiming to enhance their competitiveness and improve user experience. Predictive models with feature interpretability techniques like SHAP (SHapley Additive exPlanations), provide a powerful approach to understanding and predicting consumer behavior in online environments. Online purchase intent refers to the likelihood of a user completing a purchase based on various influencing factors, such as product features, website design, user behavior, demographic characteristics etc [1]. Numerous studies have explored the factors affecting purchase intent using machine learning and statistical models [2]. However, a key limitation in these models has been their opacity, often referred to as the ”black-box problem”. This lack of interpretability can hinder stakeholders from deriving actionable insights from predictive systems. SHAP- an explainable artificial intelligence (XAI) technique has gained prominence in recent years for its ability to interpret complex machine learning models. By attributing a consistent and theoretically grounded importance score to each feature, SHAP allows researchers [3] to discern the specific factors driving purchase intent and their relative contributions. This transparency not only enhances trust in the models but also supports data-driven decision-making in areas such as personalized marketing, recommendation systems and website optimization. This project aims to perform a feature-driven analysis of online purchase intent by integrating predictive modeling techniques with SHAP explanations. We used eight different machine learning models such as: XGBoost, SVM, Gaussian Naive Bayes, LightGBM, CATBoost, ADABoost, Random Forest, Decision Tree to build robust predictive models. CatBoost proved to be the best performing model with an accuracy of 89% and a F1-score of 67%. Then by using SHAP, we further investigates the role and importance of key features, providing actionable insights into consumer behavior patterns. By bridging the gap between predictive performance and interpretability, this research contributes to the growing field of explainable machine learning in e-commerce. Additionally, it offers practical implications for online retailers and marketers by elucidating the drivers of purchase intent and identifying areas for optimization. This study underscores the importance of both predictive accuracy and model transparency in creating effective and consumer-centric strategies in the digital marketplace.

II. Literature Review

Several machine learning techniques have been investigated for predicting customer purchasing behavior. Chaubey et al [4]. conducted a study utilizing supervised classification methods including logistic regression, decision trees, K-nearest neighbors (KNN), Naive Bayes, support vector machines (SVM), and ensemble approaches like AdaBoost and XgBoost. Among these, hybrid algorithms employing stacking, such as KnnSgd, achieved superior performance with the highest accuracy of 92.42% by effectively minimizing combined errors. This study underscores the potential of ensemble and hybrid models in improving the predictive accuracy of customer behavior analysis. Similarly, machine learning techniques are increasingly employed in the UK financial sector to enhance revenue generation and understand customer behavior. Ahmad Aldelemy and Raed A. Abd-Alhameed [5] developed a machine learning workflow aimed at achieving high classification accuracy and improved prediction confidence. Using a binary classification approach on a publicly available dataset from a Portuguese financial institution as a proof of concept. Their methodology involved data analysis, transformation and training-testing of classifiers; including Naïve Bayes, Decision Trees, Random Forests, Support Vector Machines, Logistic Regression, Artificial Neural Networks, AdaBoost, and Gradient Descent.

By employing stratified k-fold cross-validation (k=5) and assembling top-performing classifiers into a decision-making committee, they achieved over 92% accuracy with two-thirds voting confidence. The proposed workflow is simple, adaptable and suitable for practical implementation in UK banks, maintaining data privacy. Future work includes extending the approach to UK banks, reformulating the problem as multi-class classification and introducing pre-training automated steps for data analysis and transformation.

Wen, Lin, and Liu [6] further highlighted the significance of machine learning in e-commerce by addressing the challenge of predicting anonymous consumer purchase intent. They developed a machine learning model: MBT-POP, which integrates multi-behavioral trendiness (MBT) and product popularity (POP) and tested it on a large dataset comprising 33,339,730 clicks from 445,336 sessions of real e-commerce customers. Their model achieved an F1 score of 0.9031, with optimal results obtained using a sliding window of two days. Compared to previous studies, the MBT-POP model not only demonstrated superior performance but also reduced the time frame required for accurate predictions. This research emphasized that incorporating product trendiness and popularity significantly enhances the predictive performance of models, proving critical for accurately forecasting the purchase behavior of anonymous customers in e-commerce settings.

In addition, Shatabda [7] proposed a purchase intention prediction model to enable early detection of customer intent during online sessions. The approach involved feature selection, oversampling for dataset balancing, and training supervised classifiers, including SVM, random forest, decision tree, MLP and XGBoost. The XGBoost classifier outperformed other methods, achieving an auROC score of 0.937, an auPR score of 0.754 and an accuracy of 90.65%. This study demonstrated that gradient boosting methods with feature selection and balanced datasets could significantly improve purchase prediction accuracy, offering a robust approach for practical e-commerce applications.

Moreover, recent advancements in explainable AI (XAI) have enabled the interpretation of machine learning models, such as XGBoost using SHAP (SHapley Additive exPlanations). [8]This method was shown to extract spatial effects effectively, as demonstrated through simulation experiments comparing SHAP-explained XGBoost with traditional spatial models like Spatial Lag Model (SLM) and Multi-scale Geographically Weighted Regression (MGWR). An empirical example using Chicago ride-hailing data illustrated the applicability of SHAP to real-world datasets. These findings suggest that locally interpreted machine learning models are effective alternatives to traditional spatial statistical models, particularly when dealing with complex spatial and non-spatial effects. [9]This study predicts online customer purchase behavior on e-commerce platforms using deep learning techniques applied to a large dataset of over 50,000 unique web sessions, incorporating platform engagement and customer characteristics as key predictors. The deep learning approach outperformed traditional machine learning methods such as Decision Tree, Random Forest, SVM and ANN, offering insights to enhance platform design and expand academic understanding of purchase prediction.

III. Methodology

A. Dataset Description

The dataset used in this project, sourced from Kaggle, contains online shoppers' browsing behavior and purchasing patterns, with features categorized into several groups [10]. Web Page Engagement features, such as Administrative, Administrative Duration, Informational, Informational Duration, ProductRelated, and ProductRelated Duration, measure the time spent on different page types, reflecting user interest and engagement. Visitor Behavior Metrics, including BounceRates, ExitRates, and PageValues, provide insights into user actions, such as single-page visits, exit percentages, and the value of pages leading to transactions. The SpecialDay feature, ranging from 0 to 1, indicates the proximity of visits to occasions like Mother's Day or Valentine's Day, showing their potential impact on behavior. Additionally, features like Month, OperatingSystems, Browser, Region, TrafficType, VisitorType, and Weekend offer demographic, technical, and browsing context. The target variable, Revenue, is binary, indicating whether a visitor made a purchase (True) or not (False). The primary objective is to develop a model that accurately predicts Revenue based on these features.

B. Data Preprocessing

Data preprocessing is an essential step in building an effective machine learning model, as it ensures the data is clean, consistent, and ready for training. In this study, several preprocessing steps were performed to enhance the dataset's quality and compatibility. First, the dataset was thoroughly examined for missing values. Fortunately, no missing values were found in any of the features, ensuring completeness. Next, data transformation was carried out. Firstly, we dropped the duplicates rows.Then the Month column, which originally contained month names, was converted into numerical values ranging from 1 to 12 to facilitate better model interpretation and compatibility. Similarly, the VisitorType and Weekend columns, representing categorical data, were encoded into numerical codes using categorical encoding, allowing the model to utilize this information effectively. The target variable, Revenue, was converted to an integer data type (int64) to ensure consistency with machine learning algorithms. Then, class imbalance in the dataset was addressed. A significant imbalance was observed, with a higher number of instances belonging to the "False" class (no revenue) compared to the "True" class (revenue), shown in Fig 1. To resolve this issue, the Synthetic Minority Over-sampling Technique (SMOTE) was applied to the training data. SMOTE generates synthetic samples for the minority class, which helps balance the class distribution and enhances the model's ability to learn and predict the minority class accurately. These preprocessing steps ensured that the data was prepared effectively for the

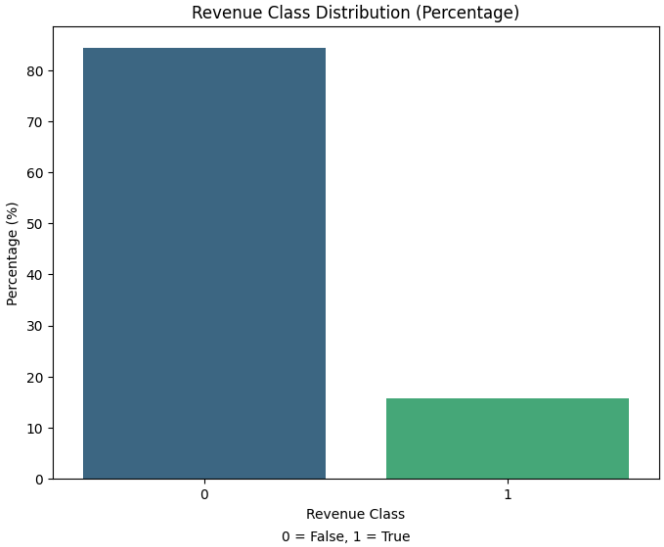

Fig. 1. Revenue Class Distributions

machine learning pipeline, improving the model's reliability and performance.

C. Learning Phase

The learning phase focuses on training and evaluating machine learning models to determine the best performing one for a specific task. First, the preprocessed dataset is split into training and testing sets using the `train_test_split` function, typically with a ratio of 70% for training and 30% for testing. This step ensures that the model is evaluated on unseen data, allowing an assessment of its generalization ability. Next, various classification algorithms are selected based on their compatibility with the task and their ability to handle the dataset's characteristics. Each selected model is then trained on the training data, where the features and target variable are used to help the model learn patterns and relationships within the data. Finally, the trained models are evaluated using the testing set and performance metrics such as accuracy, precision, recall, F1-score, and ROC-AUC. Among these, the F1-score, which balances precision and recall, is chosen as the primary metric for model selection. This metric is especially important for imbalanced datasets, as it considers both the model's ability to correctly identify positive cases and its effectiveness in avoiding the misclassification of negative cases. The following classification algorithms were employed in this study:

1) Support Vector Machine (SVM): SVM was chosen for its ability to handle high-dimensional data and complex relationships between features. The dataset used in this study has a moderate number of features, and SVM is known to perform well in such scenarios. Additionally, SVM's ability to find optimal hyperplanes for separating data points makes it a good candidate for classification tasks.

2) Decision Tree: Decision Trees were selected for their ease of interpretation and visualization. They provide clear insights into the decision-making process of the model, making it easier to understand the factors influencing predictions. This interpretability is valuable for gaining a deeper understanding of the dataset and the factors driving customer purchase decisions.

3) Random Forest: Random Forest was chosen for its robustness to noise and outliers and its ability to handle high-dimensional data. By combining multiple decision trees, Random Forest reduces the risk of overfitting and improves prediction accuracy. This makes it a suitable choice for datasets with potential noise or irrelevant features.

4) XGBoost: XGBoost was selected for its high performance and efficiency. It is known for its ability to achieve state-of-the-art results in various machine learning tasks. XGBoost's gradient boosting approach allows it to learn complex patterns in the data and make accurate predictions.

5) Gaussian Naive Bayes: Gaussian Naive Bayes was chosen for its simplicity and speed. It is a fast algorithm, making it suitable for large datasets. Despite its simplicity, Naive Bayes can be surprisingly effective in certain scenarios, especially when the independence assumption holds reasonably well.

6) LightGBM: LightGBM was selected for its efficiency and scalability. It is designed to handle large datasets with high dimensionality. LightGBM's histogram-based approach and leaf-wise tree growth strategy make it a fast and memory-efficient algorithm.

7) CatBoost: CatBoost was chosen for its ability to handle categorical features effectively. The dataset used in this study contains several categorical features, and CatBoost is known for its superior performance in such cases. It uses a novel technique to handle categorical features without requiring extensive preprocessing.

8) AdaBoost: AdaBoost was chosen for its ability to combine multiple weak learners into a strong learner. This ensemble approach can improve prediction accuracy and reduce overfitting. AdaBoost is particularly effective when used with decision trees as base learners.

D. SHAP

The best performing model was then selected and the SHAP was implemented on that particular model.This helps to interpret the model by identifying which features are most influential and how the feature values correlate with their impact.

IV. Results

Our project aimed to implement machine learning models in order to identify users online purchase intent along with using explainable AI (XAI) to provide comprehensive model analysis.

A. Model Performance comparison

We implemented eight different ML models. Among them, other than SVM and Gaussian Naive Bayes, every other model has achieved an accuracy over 80%. CatBoost has outperformed every other model with an accuracy of 89%

and a F1 score of 67%. XGBoost, LightGBM and Random Forest have also achieved an accuracy closer or similar to CatBoost. Same can be said about F1 scores too. But CatBoost was finally chosen as the best performing model based on a slight margin of F1 score. In Fig. 2 and Fig. 3, between the worst performing model and the best performing model, we can see that SVM has performed moderately in identifying class 0 but struggled with class 1, as indicated by the high false positive (829) and relatively low true positive (397) count. While CatBoost has successfully identified 2840 instances correctly for class 0 and 410 instances from class 1. In Table I, it shows the overall comparison of the models.

TABLE I
MODEL EVALUATION METRICS

ML Models	Accuracy	Precision	Recall	F1-Score	ROC-AUC
SVM	0.7258	0.3238	0.6941	0.4416	0.7919
Decision Tree	0.8528	0.5226	0.6661	0.5857	0.7767
Random Forest	0.8845	0.6097	0.7238	0.6619	0.9214
XGBoost	0.8861	0.6228	0.6871	0.6534	0.9190
Gaussian NB	0.7758	0.3829	0.7115	0.4979	0.8090
LightGBM	0.8845	0.6107	0.7185	0.6602	0.9255
CatBoost	0.8875	0.6212	0.7168	0.6656	0.9234
AdaBoost	0.8758	0.5807	0.7360	0.6492	0.9033

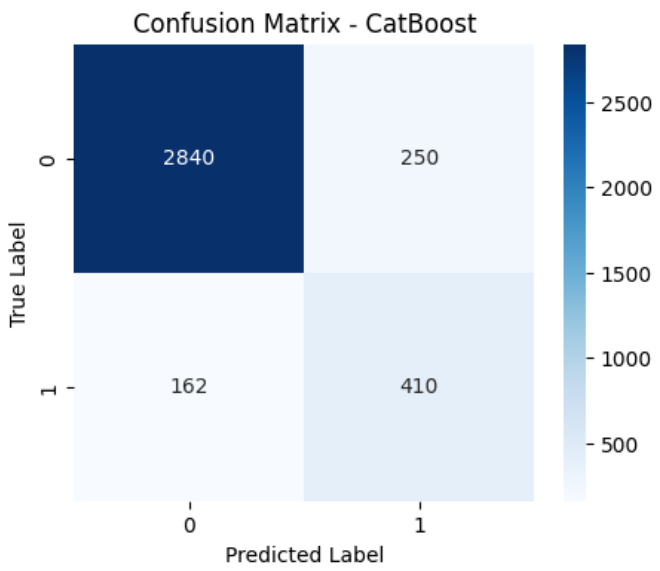

Fig. 2. Confusion Metrix - CatBoost

B. SHAP Analysis Outcome

Fig. 4 shows the SHAP feature importance plot. It shows that PageValues has the highest impact on the model prediction followed by the Month. Administrative Duration, exit rates and visitor type also have a moderate importance. While features like Traffic Type, Special Day and Browser have minimal impact on the model predictions. A violin diagram in Fig. 5 has been plotted to showcase the SHAP summary plot of Fig. 4. It shows that page values largely influence the output with high values strongly pushing the predictions positively. Which

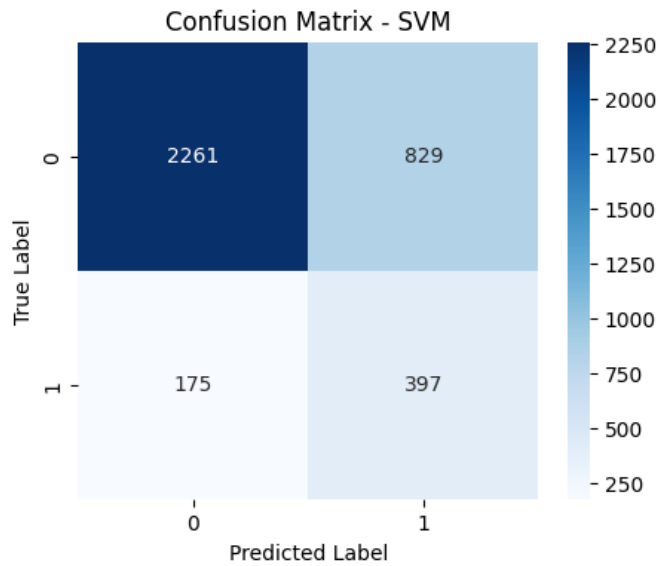

Fig. 3. Confusion Metrix - SVM

indicates the user will purchase while the opposite side means the user is unlikely to purchase. And bounce rates and exit rates often lead toward negative revenue when the values are high. Same goes for the higher value of Administrative.

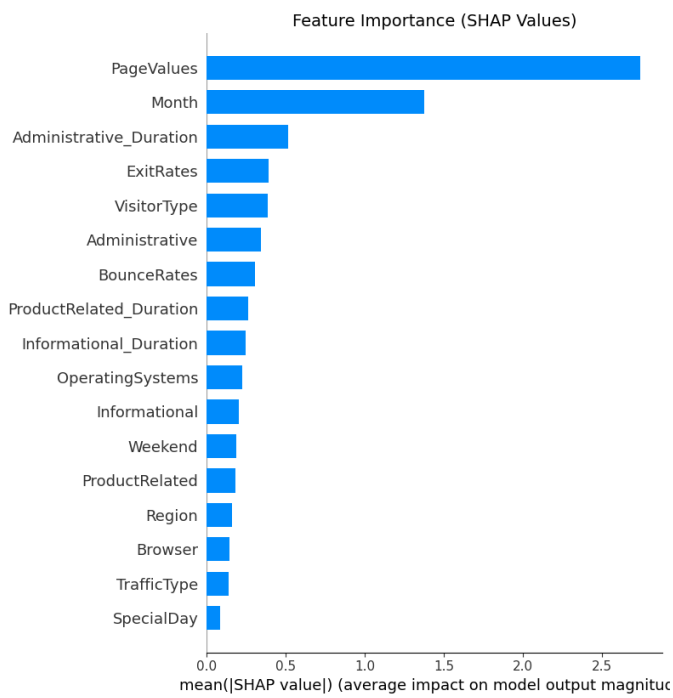

Fig. 4. SHAP feature importance

V. CONCLUSION

This study not only explored different machine learning models but also tried to leverage explainable Ai(XAI) model such as SHAP, in order to present a feature-driven analysis of

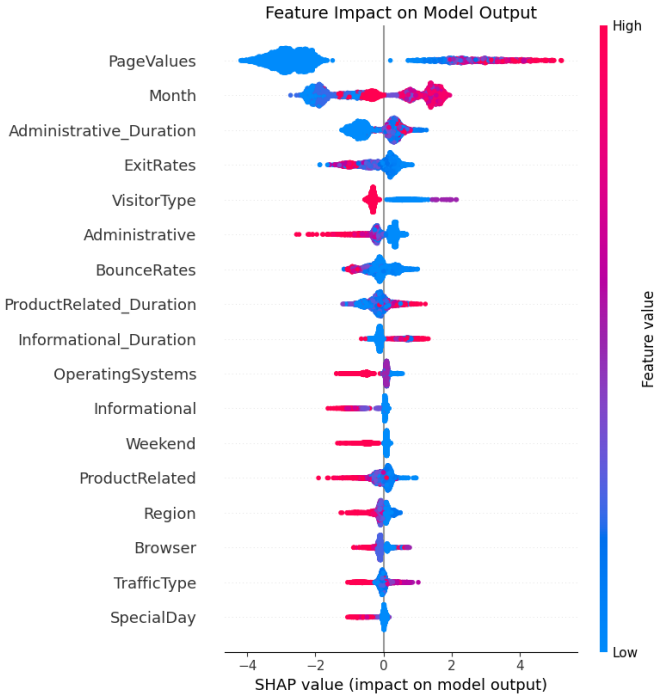

Fig. 5. SHAP summary plot

the purchasing intent of the online shoppers. Despite having a moderate F1-score, our model has performed well with accuracy. In future, works can be done with a focus on using more advanced XAI models like LIME and ALE and exploring DL models to provide more insights and achieving higher accuracy.

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